

SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 1.2: Chemicals of Life — Teacher Reference (Pre-filled)

SUMMARY TABLE: BIOLOGY GRADE 10	
Sub-Strand	Sub-Strand 1.2: Chemicals of Life
Driving Question	DRIVING QUESTION: Why do two children in Kisumu eating the same amount of food every day end up with completely different bodies — and what does that tell us about what food is really made of?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 Predict: What Is Food Really Made Of?	Students observed two photographs of children from Kisumu — one with signs of kwashiorkor (swollen belly, thin limbs, discoloured hair, fatigue) and one healthy and energetic — and were told both children eat similar quantities of food daily, with Child A eating ugali and Child B eating githeri.	Students learned that food is not simply 'food' — it is composed of different chemical substances (carbohydrates, proteins, lipids, vitamins, mineral salts, water), and that two foods can look similar in quantity but be chemically very different. They were introduced to the names of the six major chemical families of life and matched them to familiar Kenyan foods in a first-draft card sort.	Students cannot yet fully explain why the two children have different bodies, because they do not yet know which specific chemicals are present in ugali versus githeri, what each chemical does in the body, or how to test for their presence. Their first explanatory models show a beginning of cause-effect thinking (missing something in ugali → sick body) but lack the chemical detail that food tests in upcoming lessons will provide. The driving question remains open: Why do two children in Kisumu eating the same amount of food every day end up with completely different bodies — and what does that tell us about what food is really made of?
Lesson 2 What Is Food Really Made Of? Introducing the Six Chemicals of Life	Learners observed two photographs of children in Kisumu — one with kwashiorkor (eating ugali only) and one healthy (eating githeri) — and sorted laminated cards of 10 Kenyan foods (ugali, githeri, omena, sukuma wiki, avocado, milk, mango, groundnuts, eggs, white sugar) by chemical composition using percentage data printed on the reverse of each card.	Learners learned that all food is composed of six major chemicals of life — carbohydrates, proteins, lipids, nucleic acids, vitamins/minerals, and water — each with a distinct role in the body. They learned that ugali is almost entirely carbohydrate (energy) with very little protein (building/repair), while githeri combines carbohydrate (maize) with significant	Learners explained, through a written Cause and Effect response and a first-draft molecular model, why the child eating ugali alone developed kwashiorkor: the body received sufficient energy (carbohydrates) but insufficient protein, so it could not build or repair muscle tissue, produce enzymes and hormones, or maintain normal fluid balance — causing the visible symptoms of

		protein (beans), explaining why the same quantity of food produces dramatically different health outcomes.	swollen belly, thin limbs, discoloured hair, and fatigue seen in the photograph.
Lesson 3 Observe (Part 2): Lipids and Vitamins — Structure, Function, and Food Tests	Learners performed the translucent spot test and ethanol emulsion test on 8 Kenyan food samples (ugali, groundnuts, avocado, githeri, coconut flesh, whole milk, sukuma wiki, mandazi) and recorded which foods contain lipids. Learners observed that ugali tested negative for lipids in both tests, while groundnuts, avocado, coconut, and milk tested strongly positive. Learners also read and annotated a structured text on fat-soluble and water-soluble vitamins, identifying their roles and Kenyan food sources.	Learners learned that lipids are composed of glycerol and fatty acids and serve as long-term energy stores, cell membrane components, and carriers for fat-soluble vitamins (A, D, E, K). They learned that vitamins are micronutrients needed in small quantities but critical for metabolic processes, immunity, and organ function. Critically, they learned that fat-soluble vitamins cannot be absorbed without dietary fat — meaning a fat-free diet causes both direct lipid deficiency AND indirect vitamin deficiency. Ugali's chemical profile now includes: rich in carbohydrates, low in protein (Lesson 2), and absent in lipids and fat-soluble vitamins (Lesson 3).	Learners constructed CER (Claim–Evidence–Reasoning) paragraphs arguing that the Kisumu child eating only ugali is deficient in lipids and fat-soluble vitamins in addition to protein. They explained that this deficiency would cause multiple overlapping health effects — not only the swollen belly and thin limbs of kwashiorkor (protein deficiency, Lesson 2), but potentially also night blindness (Vitamin A), weak bones (Vitamin D), and impaired immunity. Learners updated their Cumulative Chemical Model of Food to include lipids and vitamins, connected by a structural arrow showing fat-soluble vitamin absorption dependency, and clearly labelled all chemical gaps in ugali.
Lesson 4 Building Life's Molecules: Structure of Carbohydrates and Proteins	Learners constructed annotated diagrams of carbohydrate and protein molecular structures from guided reading passages, built physical models of monosaccharides, disaccharides, polysaccharides, and dipeptides using bottle tops and straws, and mapped Kenyan foods (ugali, githeri, nyama choma, sugarcane juice, chai) to their correct molecular classification (monosaccharide, disaccharide, polysaccharide, protein).	Carbohydrates are made of monosaccharide monomers (e.g. glucose) joined by glycosidic bonds to form disaccharides (e.g. sucrose in chai) and polysaccharides (e.g. starch in ugali, glycogen stored in muscle/liver). Proteins are made of amino acid monomers joined by peptide bonds; amino acids contain nitrogen (N) in addition to C, H, and O — an element absent in carbohydrates. This elemental and structural difference means starch can supply energy but cannot provide the building blocks for muscle, blood proteins (haemoglobin), or albumin. A diet of ugali alone provides no amino acids, so the body cannot build or repair proteins — explaining the swollen belly (loss of albumin), thin limbs (loss of muscle protein), and discoloured hair (loss of keratin) seen in the Kisumu child with kwashiorkor.	The two Kisumu children eating similar quantities of food have completely different bodies because the MOLECULAR COMPOSITION of their food differs, not just the amount. Ugali is almost entirely starch — a polysaccharide composed only of glucose monomers containing C, H, and O. Githeri adds beans, which are rich in protein — polymers of amino acids containing N. Without amino acids, the child's body cannot synthesise the proteins essential for tissue building, immune function, and fluid balance. The anchoring phenomenon (kwashiorkor) is therefore a direct consequence of molecular absence: no amino acids in the diet → no protein synthesis → visible, devastating loss of body structure and function. Food quantity is irrelevant if the molecular composition is incomplete.
Lesson 5 Sensemaking: Connecting the Chemistry of Food to Health Outcomes in	In Lesson 4, learners observed that iodine turned blue-black in ugali (confirming starch) but the Biuret test on ugali extract showed no purple colour (no protein), while the Biuret test on beans extract turned	Learners learned that kwashiorkor is caused specifically by protein deficiency — not by insufficient food quantity. The child eating ugali alone receives carbohydrates for energy but no amino acids, so the body	Learners constructed evidence-based scientific explanations linking their food test results and molecular structure knowledge to the anchoring phenomenon. The explanation demonstrates that food is not

Kisumu	purple (protein present). Food test data showed githeri contains both starch and protein, while ugali contains starch only.	cannot synthesise albumin (causing oedema/swollen belly), keratin (causing hair discolouration), or muscle proteins (causing thin limbs). Githeri provides essential amino acids from beans that complement the maize carbohydrate, explaining the healthy child's better health outcome. Five deficiency diseases were matched to their missing chemical of life: kwashiorkor (protein), marasmus (total caloric energy including protein and carbohydrate), scurvy (vitamin C), rickets (vitamin D and calcium), anaemia (iron).	uniform — it is a mixture of specific chemicals of life, each with irreplaceable structural and functional roles in the body. The quantity of food matters less than its chemical composition, which is why two children eating the same amount of food can have completely different bodies and health outcomes.
Lesson 6 Building Our Chemicals of Life Model: Consolidating the Science Behind Two Children in Kisumu	Learners observed their own Lesson 1 prediction cards alongside the completed group visual models showing all six chemicals of life with their structures, food sources, functions and deficiency consequences. Learners also observed the Kisumu Menu Evaluation card representing a realistic Kenyan daily diet.	Learners learned that the six chemicals of life — carbohydrates, proteins, lipids, nucleic acids, vitamins and minerals, and water — each have distinct molecular structures, specific biological functions and identifiable Kenyan food sources. The absence of even one chemical causes a specific, predictable and often visible health consequence. The child eating ugali develops kwashiorkor because maize starch provides carbohydrates but no essential amino acids, so the body cannot synthesise albumin, keratin or muscle proteins. The githeri child remains healthy because beans supply essential amino acids that complement the maize carbohydrate.	Learners explained that food is not interchangeable by quantity alone — its chemical composition determines its biological effect. A diet sufficient in carbohydrates can still cause severe disease if proteins are absent, because each chemical of life performs functions that no other chemical can substitute. This understanding explains the difference between the two Kisumu children and connects molecular biochemistry to community health outcomes across Kenya.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

Production & Layout: Educational design and curriculum sequencing by SeaVuria. Technical layout, structural formatting, and digital optimization/implementation executed in partnership with ARES Education (registered 501(c)(3) in the United States and Canada (as Afretech Aid) and registered NGO in Kenya), with lesson drafting and compilation assisted by AI (Claude, Anthropic).

Licensing & Copyright: © 2026 SeaVuria and ARES. This work is licensed under a [Creative Commons Attribution-NonCommercial 4.0 International License \(CC BY-NC 4.0\)](https://creativecommons.org/licenses/by-nc/4.0/). To view a copy of this license, visit <https://creativecommons.org/licenses/by-nc/4.0/>. You are free to copy, redistribute, and adapt this material in any medium or format, provided appropriate credit is given and the material is not used for commercial purposes. KICD learning outcomes and curriculum designations reproduced here for alignment purposes remain the property of KICD and are not covered by this license. Linked external resources (e.g. CK-12, Khan Academy, PhET) are subject to their own licenses.